Types of computers:

- There are two types of computers

Fixed Program

- Performs only a specific task
- Cannot be re-programmed to do other tasks

Stored program

- Performs different tasks
- Can be reprogrammed

Von Nuemann Architecture

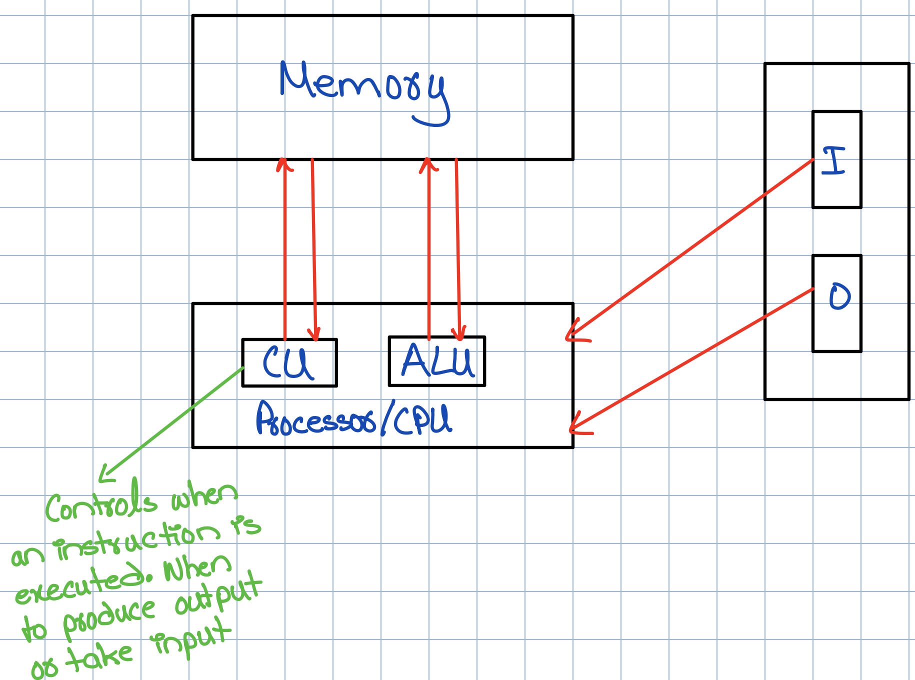
- Based on three basic characteristics:

1. It has four basic components:

    1. Memory
    2. ALU
    3. Control Unit
    4. Input/Output System (I/O)

    (2. ALU and 3. Control Unit: combined into CPU)

2. Program is stored in memory during execution
3. Program instructions are executed sequentially



Memory subsystem:

- Consists of memory cells of fixed size.
- Each cell has a address associated with it.

Words:

- A Basic unit that is stored/retreived from RAM
- Can be of varying size.
- Usually 8 bits or 1 byte = 1 word

- RAM size is multiples of 8

- N is the number of address bits, 2^N is total number of addressable cells.

Example,

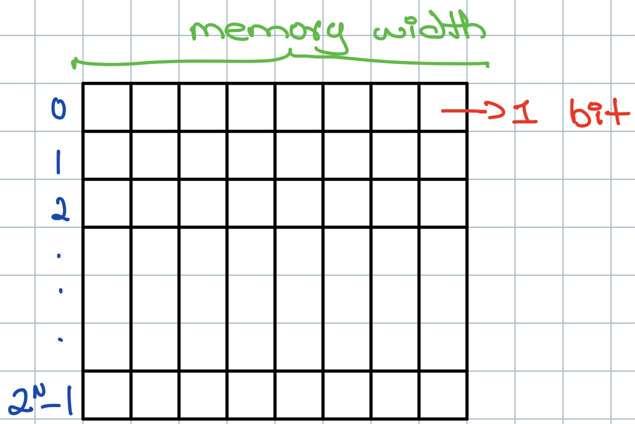
$N = 3$, $2^3 = 8$

0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

8 cells with 3 bit unique address

Also called address space

- Control Unit tells which address corresponds to which cell.



Operations on memory

1. Fetch/Read
   - Copy contents of a memory cell with specified address

2. Write/Store
   - Store specified value into memory cell specified by address

Input/Output Subsystem:

- Handles device that allow the computer system to:
  - Communicate and interact with the outside world
  - Store information (HDDs, CD-ROMs)

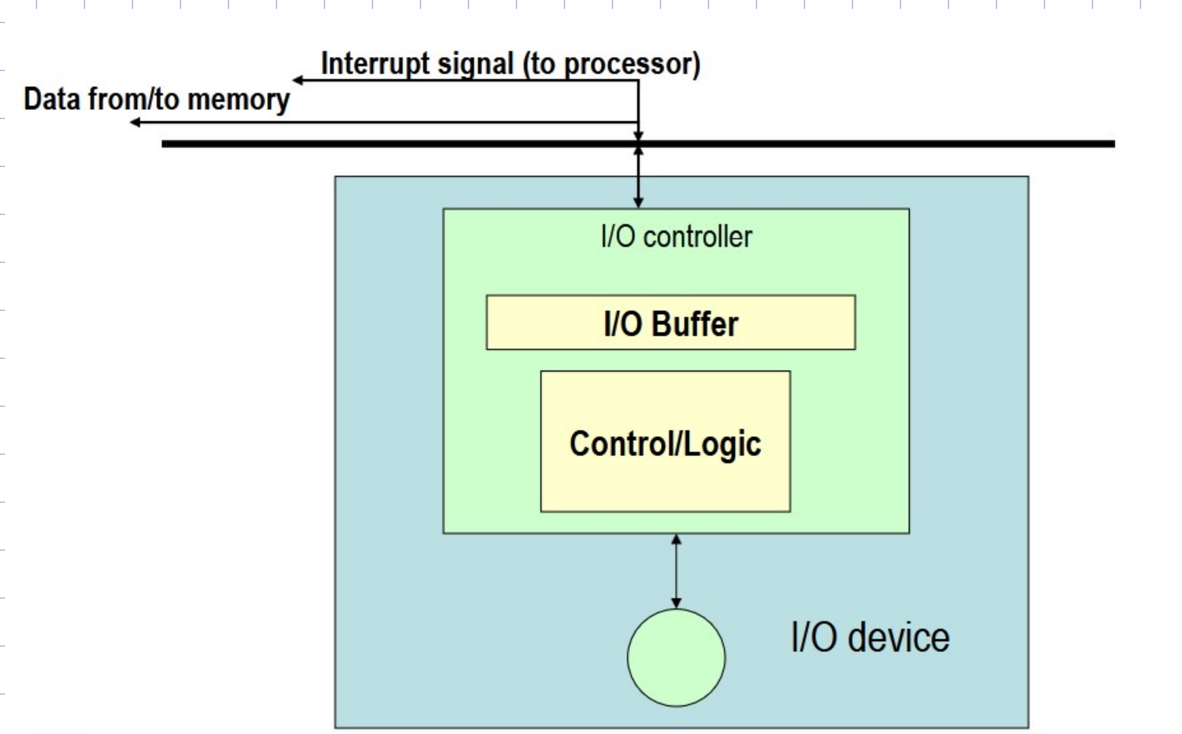
- The RAM is very fast compared to an IO device

  RAM: 50 ns   Harddrive: 10ms

Solution:
- I/O controller, a special processor that balances the gap between I/O device & processor.

- Data is transferred b/w RAM & I/O buffer

- Processor is free to do something else while the I/O controller reads or writes to or from device to I/O device.



ALU Subsystem:

↳ Performs mathematical & logic operations
↳ Integrated into CPU

↳ Consist of:
  ↳ Circuits to do the arithmetic/logic operations
  ↳ Registers to store intermediate results

Control Unit:

↳ Execute program repeatedly

1. Fetch from memory the next instruction to be executed
2. Decode it, that is, determine what is to be done
3. Execute it by issuing the appropriate signals to the ALU, memory and I/O subsystems
4. Continues until HALT (end) instruction

Bus:

↳ Communication line that is used to transfer from one part of computer to another.

1. Address bus → Communicates address to diff components
2. Data bus → Communicates data to diff components
3. Control bus → Sends control signals

Machine level instructions:

op code   Address 1   Address 2
≤ 8 bits

Program Counter

↳ PC is set to the address where the first program instruction is stored
↳ Repeat
  ↳ Fetch instruction
  ↳ Decode instruction
  ↳ Execute instruction
End

Machine Language Instructions

↳ Consist of

operation code   Address 1   Address 2

Example,

Let 1001 is op code for ADD,
Addresses 99 & 100

So,

00001001   0000000001100011   0000000001100100